Sanath Kumar Coonani

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Summary

ML Engineering student with OCR research experience at C-DAC, taking models from data pipeline and training to Dockerised inference APIs across deep learning, reinforcement learning, and explainable ML.

Technical Skills

- **ML / Deep Learning**: PyTorch, Scikit-learn, LightGBM, Transformers, CNNs, LSTMs, Reinforcement Learning (DQN, MARL), Fine-tuning, SHAP
- **Computer Vision & OCR**: OpenCV, YOLOv8, PaddleOCR (PP-OCRv5), SwinV2
- **Data**: Pandas, NumPy, Matplotlib, Feature Engineering, Imbalanced Data, Model Evaluation
- **MLOps & Deployment**: Docker, Flask, REST APIs, GPU Training (SLURM), AWS, Azure, Git
- **Languages & Web**: Python, C++, Java, JavaScript; Node.js, Next.js, Django

Experience

- **AI/ML Research Intern – Centre for Development of Advanced Computing (C-DAC)** *May 2026 – July 2026*
  Tamil OCR research, IndicOCR initiative *Chennai, On-site*
  - Shortened OCR target sequences by **40%** (7.94 → 4.76 tokens per word) by designing **akshara-level tokenization** for Tamil using **Unicode grapheme clusters**, replacing code-point tokenization.
  - Reached **93.6% validation sequence accuracy** (0.986 normalised edit distance) by integrating the tokenizer into **PaddleOCR**'s CTC and NRTR paths and **fine-tuning PP-OCRv5** on a **SLURM GPU cluster**.
  - Built a word recognizer from scratch (**SwinV2** backbone, **BiLSTM**, CTC / **Transformer** decoders); traced its errors to dropped combining marks through **error analysis**, which motivated the tokenizer redesign.
  - Built a **355-token akshara dictionary** covering **99.8%** of 80K labels by auditing and NFC-normalising the corpus.
- **ML Intern – Temperate Technologies** *June 2025 – August 2025*
  Cold-storage climate control *On-site*
  - Built a **Scikit-learn** model to predict environmental suitability for crop storage from cleaned sensor data using **Pandas**.
  - Trained a **reinforcement learning control policy** to regulate temperature and humidity under energy constraints, evaluating stability with **NumPy** and **Matplotlib**.
  - Served real-time inference and automated control through a containerised **Flask REST API** on **Docker**, processing live sensor data.
- **Software Development Intern – Client Project *(Live Website)*** *December 2025 – February 2026*
  Next.js, Node.js, TailwindCSS *Hybrid*
  - Built and deployed a production website serving **5,000+ monthly users** with **Next.js** and **Node.js** REST APIs.
  - Improved page load speed by **40%** through asset optimisation and performance tuning.

Projects

- **Customer Behaviour Analysis (ShopLens) *(GitHub)*** 2026
  E-commerce friction detection & recovery – TCS Hackathon *Python, LightGBM, SHAP*
  - Identified the correct cause for **83.7%** of friction-driven cart abandonments (**92.2%** top-2) on **47.5K** held-out sessions by training a class-weighted **LightGBM** model over 10 friction types.
  - Flagged at-risk shoppers mid-session with **0.94 AUC** after 20+ events (0.86 overall) and **under 1 ms** scoring by training a drop-risk model on partial clickstream sessions.
  - Simulated **500K sessions** (**5.7M** clicks, **64.7%** abandonment); **SHAP** explanations drive live shopper nudges and a seller dashboard.
- **Traffic Signal Optimization with Multi-Agent RL *(GitHub)*** 2025
  Decentralised traffic control for Indian intersections *Python, PyTorch, SUMO*
  - Cut average wait per vehicle by **57.7%** (14.7s → 6.2s) and total wait time by **79.8%** versus fixed-timer signals by training independent **DQN** agents per intersection.
  - Added a hierarchical master controller that creates green corridors, bringing emergency-vehicle wait to **near zero**.
- **Geometry Dash AI Agent *(GitHub)*** 2025
  Vision-based deep reinforcement learning *Python, PyTorch, OpenCV*
  - Improved average survival time **4.5x** over a random baseline by training a **CNN-based DQN** on **100K+ frames** with experience replay (50K buffer), epsilon decay, and target networks.
  - Built the RL environment around real-time **screen capture** and **automated input** at **30+ FPS** with **OpenCV**.

Education

- **University of Hyderabad** 2023 – 2028
  Integrated Masters (M.Tech) in Computer Science and Engineering *Hyderabad, India*

Leadership

- **Lead Organizer, Codosapiens Coding Club** (University of Hyderabad, 2025 – Present): revived the club, running coding contests, workshops, and hackathons.